O-C-426d
AUGUST 19, 1957
SUPERSEDING
Int. Fed. Spec. O-C-00426c(CSA-FSS)
July 18, 1956 and
Fed. Spec. O-C-426b
August 21, 1953

FEDERAL SPECIFICATION

CLEANING COMPOUND, TOILET BOWL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — This specification covers a cleaning compound essentially made of sodium bisulfate for use in removing stains and discolorations from toilet bowls.

1.2 Classification.

1.2.1 *Types.* — Cleaning compound, toilet bowl, covered by this specification shall be of the following types, as specified:

Type I.—Crystalline or granular.

Type II.—Globular.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

NN-P-515—Plywood, Container Grade.
RR-S-366—Sieves, Standard for Testing Purposes.
PPP-B-585—Boxes; Wood, Wirebound.
PPP-B-591—Boxes, Fiberboard, Wood-Cleated.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Boxes, Fiber.

Federal Standards:

Federal Standard No. 102 — Preservation, Packaging, and Packing Levels.
Federal Standard No. 123—Marking for Domestic Shipment (Civilian Agencies).
Federal Test Method Standard No. 536—Soap and Soap-Products (Including Synthetic Detergents); Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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FSC Class 7930

Military Specifications:

MIL–A–101 — Adhesive, Water-Resistant, for Sealing Fiberboard Boxes.

MIL–B–138—Boxes, Wood, Fiberboard-Lined for Overseas Shipment.

MIL–B–10377—Box, Wood, Cleated, Veneer Paper Overlaid.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military specifications and standards, required by contractors in connection with specific procurement functions, should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. — The following publications, of the issues in effect on date of invitation for bids, forms a part of this specification:

Federal Caustic Poison Act

(Copies of the Government publication referenced above may be obtained upon application, accompanied by money order, coupon, or cash to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Prices may be obtained from the Superintendent of Documents.)

Consolidated Freight Classification Rules and Container Specifications.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Type I.*—Type I cleaning compound shall be in the form of colorless to brick red crystals or granules of sodium bisulfate, and shall be without hard caking in the container, when received.

3.1.2 *Type II.*—Type II cleaning compound shall be in the form of colorless to light straw yellow globules of sodium bisulfate, and shall be without hard caking in the container, when received.

3.2 Acidity (H_2SO_4).—The material shall contain a minimum of 30 percent sulfuric acid when tested as specified in 4.4.2.

3.3 Particle size.—The material shall conform to table I when tested as specified in 4.4.3.

TABLE I.—*Particle size*

Sieve number	Type I percent retained		Type II percent retained	
	Minimum	Maximum	Minimum	Maximum
6	--	0	--	--
8	--	--	0	5
20	--	--	80	--
30	90	--	--	--
40	--	--	90	--

3.4 Moisture content (type II).—The moisture content of the material shall be a maximum of 1.5 percent when tested as specified in 4.4.4.

3.5 Iron content (type II).—The iron content of the material shall be a maximum of 0.15 percent when tested as specified in 4.4.5.

3.6 Percent insoluble matter (type II). — The material shall contain not more than 0.25 percent insoluble matter when tested as specified in 4.4.6.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.*—For purposes of sampling, a lot shall consist of all cleaning compound (toilet bowl) of the same type manufactured as one batch and offered for delivery at one time.

4.1.2 *Sampling for inspection of filled containers.*— A random sample of filled containers shall be taken from each lot in accordance with Military Standard MIL–STD–105 at Inspection Level I and Acceptable Quality Level (AQL) = 2.5 percent defective to verify compliance with all stipulations of this speci-

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fication regarding fill, closure, marking, and other requirements not involving tests.

4.1.2.1 *Standard MIL–STD–105.* — When Military Standard MIL–STD–105 specifies an action by the Government, it shall, at the option of the Government, be performed either by the Government or by the contractor under the supervision of the Government inspector.

4.1.3 *Sampling for tests.*—From each inspection lot the Government inspector shall take four containers at random. Two containers shall be forwarded to the testing laboratory designated by the bureau or agency concerned, one shall be delivered to the contractor, and one shall be held by the inspector to be used in case of dispute.

4.2 Inspection.

4.2.1 Inspection may be made at the site of manufacture, at any suitable point in transit, or after delivery to the point of destination, as specified in the contract or order.

4.2.2 *Inspection of filled containers.*—Each sample filled container selected in accordance with 4.1.2 shall be examined for defects of the container and the closure, for evidence of leakage, and for unsatisfactory markings. Each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample having one or more defects or under required fill shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Military Standard MIL–STD–105 the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for acceptance tests provided the contractor has removed or repaired all nonconforming containers.

4.3 Lot acceptance test.—The sample specimens selected in accordance with 4.1.3 shall be subjected separately to the tests specified in 4.4. If either specimen fails in one or more of these tests the lot shall be rejected. Rejected lots may be resubmitted for acceptance tests provided the contractor has removed or reworked all nonconforming products.

4.4 Test procedures.

4.4.1 Visual examination. — The sample shall be examined to determine if it is free from hard lumps in the container, and if it is in granular form in the case of type I and globular form in the case of type II.

4.4.2 *Acidity* (H_2SO_4). — Weigh quickly 10 gm. ± 0.01 gm. of the sample, dissolve and dilute to 500 ml. with distilled water; transfer a 50-ml. aliquot portion to a 400-ml. beaker, add about 200 ml. of distilled water and 2 drops of a 1 percent phenolphthalein indicator solution. Titrate to a distinct pink end point with a 0.2 *N* sodium hydroxide solution.

Calculation:

Percent H_2SO_4 = ml. 0.2 *N* NaOH × 0.98.

Note.—Weight of sodium bisulfate equals weight of sulfuric acid times 2.448.

4.4.3 *Particle size.*—Sieving tests shall be made in accordance with applicable provisions of method 2101 sieve tests of Fed. Test Method Std. No. 536.

4.4.4 *Moisture content.* — Place approximately 15 grams of the sample in a tared covered Petri dish and weigh on an analytical balance. Set the Petri dish in its cover and dry overnight in an electric oven at 105°C. for 16 hours. Cool in a desiccator and reweigh. Calculate as follows:

$$\frac{\text{Loss in weight} \times 100}{\text{Weight of sample}} = \text{percent } H_2O$$

4.4.5 *Iron content.*—Dissolve 5 grams of the sample in 60 ml. of water and 20 ml. of hydrochloric acid and boil gently for 10 minutes. Cool and dilute to 100 ml. Dilute 10 ml. of this solution to 50 ml. and add 30 to 50 mg. of ammonium persulfate crystals and 3 ml. of ammonium thiocyanate reagent solution. Any red color should not exceed that produced by 0.75 mg. of iron in an equal volume

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of solution containing the quantities of reagents used in this test.

4.4.6 *Percent insoluble matter.* — Accurately weigh 50 grams of the sample and dissolve in 300 ml. of hot distilled water in a 600-ml. beaker. Allow the insoluble matter to settle and filter by decanting through a tared weighed Gooch crucible. Wash the insoluble matter into the crucible with more hot water. When the insoluble matter is completely filtered, place the crucible in an electric oven at 100° to 110°C. for 1 hour. Cool in a desiccator and reweigh. Calculate as follows:

$$\frac{\text{Weight of residue} \times 100}{\text{Weight of sample}} = \text{percent insoluble matter}$$

5. PREPARATION FOR DELIVERY

The definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Level A.*—Cleaning compound, toilet bowl, shall be packaged in round metal cans containing not less than 20-ounces (net weight), made throughout of bonderized black plate. Cans shall be multiple friction plug style with double-seamed ring and bottom, size approximately 3 1/16 inches in diameter by 4 13/16 inches in height with a 1 1/2 inch or larger top ring opening. Plug shall be curled edge, deep, tight fitting and well seated. Cans shall be lithographed and varnished, or lacquered over-all outside with corrosion-resisting materials, and enameled throughout the inside with an enamel suitable for the product.

5.1.1.1 Alternatively, cans with pour-spouts, completely lithographed, varnished or lacquered, may be furnished in lieu of friction style cans.

5.1.2 *Level C.*—The cleaner shall be packaged in quantities of not less than 20 ounces (net weight) in containers conforming to the manufacturer's practice.

5.2 Packing.

5.2.1 *Level A.*—The cleaner, packaged in accordance with level A or C as specified (see 6.2), shall be packed twenty-four cans each on end in fiberboard boxes conforming to Federal Specification PPP–B–636, type I or II, class 2, style RSC, compliance symbol V3c or V3s, or when specified, packed in exterior type shipping containers meeting Federal Specification PPP–B–591 (overseas type); PPP–B–621, class 2; MIL–B–138, PPP–B–585, class 2 or 3; PPP–B–601 (overseas type); or MIL–B–10377, style A or B. As far as practical, exterior shipping containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required and contain identical quantities. Unless otherwise specified, the gross weight of each pack shall be limited to approximately 200 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. Plywood, when used, shall conform to Federal Specification NN–P–515, type II, class 2.

5.2.2 *Level B.*—Cleaning compound packaged in accordance with level A or C as specified (see 6.2), shall be packed twenty-four cans each on end in a corrugated or solid fiber box conforming to Federal Specification PPP–B–636, type I or II, class 1, style RSC. The fiberboard shall have a bursting strength of not less than 275 p.s.i. All flaps of the fiber containers shall be securely closed with an adhesive conforming to Military Specification MIL–A–101, by application throughout the entire area of contact between the flaps.

5.2.3 *Level C.*—Cleaning compound packaged in accordance with level A or C as specified (see 6.2) shall be packed in containers in a manner to insure safe delivery and acceptance at destination. Containers shall comply with the Consolidated Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

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5.3 **Marking.**

5.3.1 *Nonmilitary agencies.*

5.3.1.1 *Unit issue package.*—Unless otherwise specified, each can or issue package shall be marked in accordance with Federal Standard No. 123. The packages shall be marked in accordance with the provisions of the Federal Caustic Poison Act, and regulations thereunder. In addition, each package shall bear the following label:

SODIUM BISULFATE

DANGER! Causes severe burns. Do not get in eyes, on skin, on clothing. In case of contact, immediately flush skin or eyes with plenty of water for at least 15 minutes; for eyes, get medical attention.

POISON

If taken internally, drink one (1) teaspoon or more of magnesia, chalk, wall plaster, or soap softened with water, milk, mucilage, or raw egg white. Get medical attention.

DIRECTIONS FOR USE

First flush the toilet; then shake the toilet-bowl cleaner into the water and around the sides of the bowl above the water line. Let stand for a few minutes; remove the discoloration with a toilet bowl brush or swab, then flush the toilet. When the bowl is badly discolored, repeat the operation, allowing the solution to remain in the bowl 15 minutes or longer.

5.3.1.1.1 The following warning shall appear on the side of the container in letters not less than ⅛ inch high:

DO NOT USE ON BATHTUBS, WASHSTANDS, SINKS, OR OTHER METAL OR ENAMELED FIXTURES.

5.3.1.2 *Shipping container.*—Unless otherwise specified, shipping containers shall be marked in accordance with Federal Specification PPP–P–31.

5.3.2 *Military agencies.*—In addition to the marking required in 5.3.1.1 and 5.3.1.1.1, shipping containers shall be marked in accordance with the requirements of Military Standard MIL–STD–129.

6. NOTES

6.1 Material should be purchased by net weight.

6.2 **Ordering data.** — Purchasers should specify the title, number, and date of this specification, the size and weight of sample unit, if required, and should exercise any desired options offered herein. (See 1.2 and section 5).

6.3 **Transportation description.**—Transportation description applicable to these items is:

Cleaning compound, not otherwise indexed by name, granular or powder.
Carload weight 36,000 pounds.
Truckload weight 36,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—Q M
Navy—Sh MC
Air Force.

Copies of this specification may be purchased for .05 cents each.

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